SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, TIRUCHIRAPPALLI

Department of Artificial Intelligence and Machine Learning

21CSC406J - NEURAL NETWORKS AND DEEP LEARNING

Challenge Based Learning (CBL)

PS-32 — Casting Defect Inspection Using Deep Learning



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Abstract

Casting defects in submersible pump impellers can affect product quality and may be difficult to identify consistently through manual visual inspection. This work develops a binary image-classification system to automatically classify casting images as **Defective** or **OK**. A transfer-learning approach using **MobileNetV2**, pretrained on ImageNet, was implemented using TensorFlow/Keras. The dataset contains 7,348 casting images. Images were resized to 160×160 pixels and normalized, with data augmentation applied during training. Since missing a defective casting is more costly than generating a false alarm, the decision threshold was selected using the validation set with an F2-oriented criterion. A threshold of **0.30** was selected, resulting in **99.78% recall for the defective class** on the test set, with only one defective casting missed.

1. Problem Analysis

The objective is to replace or assist manual visual inspection of submersible pump impeller castings. Each top-view casting image belongs to one of two classes: **Defective** or **OK**. Defects may be subtle and difficult to identify consistently through manual inspection.

The main risk is a **false negative**, where a defective casting is incorrectly classified as OK. This is more costly than a false positive because a defective component could pass inspection. Therefore, the system prioritizes **high recall for the Defective class**.

The dataset contains approximately 7,300 grayscale casting images according to the problem statement; the provided dataset contained **7,348 images** in total.

2. Dataset and Preprocessing

The provided dataset was organized into train and test folders, each containing def_front and ok_front classes.

Dataset	Defective	OK	Total
Training	3,758	2,875	6,633
Testing	453	262	715
Total	**4,211**	**3,137**	**7,348**

The original training data was divided into **85% training and 15% validation** using stratified splitting and a fixed random seed of 42. Images were resized from **300×300 to 160×160**, converted to RGB, and normalized to the range [0,1].

Data augmentation included random rotation, zoom, and contrast adjustment.

Figure 1 — Class Distribution

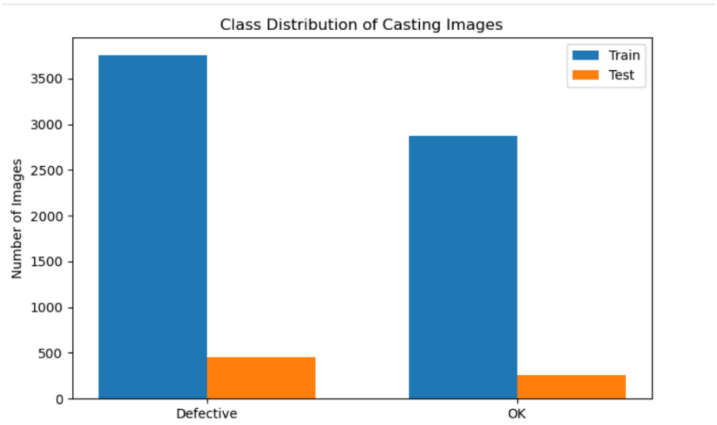


Figure 1. Distribution of Defective and OK casting images.

Figure 2 — Sample Casting Images

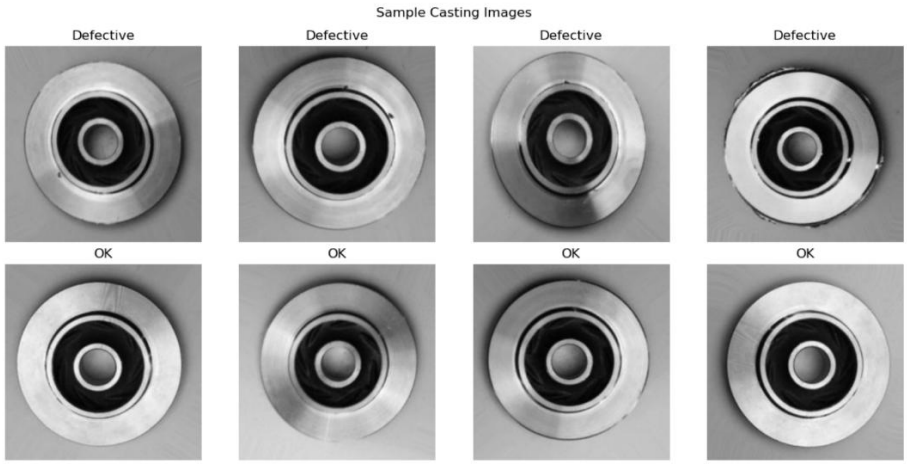


Figure 2. Representative casting images from the two classes.

3. Coding / Proposed Solution

A transfer-learning CNN classifier was developed using **MobileNetV2**. The pretrained convolutional base was frozen and a new classification head was added for binary classification.

Key Implementation Code

```
base_model = MobileNetV2(
    input_shape=(160, 160, 3),
    include_top=False,
    weights="imagenet"
)

base_model.trainable = False

inputs = layers.Input(shape=(160, 160, 3))
```

```
x = layers.RandomRotation(0.05)(inputs)
x = layers.RandomZoom(0.10)(x)
x = layers.RandomContrast(0.10)(x)

x = layers.Rescaling(2.0, offset=-1.0)(x)
x = base_model(x, training=False)

x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.30)(x)

outputs = layers.Dense(1, activation="sigmoid")(x)

model = Model(inputs, outputs)
```

The model was trained using **Adam optimization**, binary cross-entropy loss, class weighting, early stopping, and learning-rate reduction. Training was performed for a maximum of 8 epochs, with the best validation model retained.

The model achieved a validation accuracy of approximately **98.59%** at the end of training.

Figure 3 — Training Performance

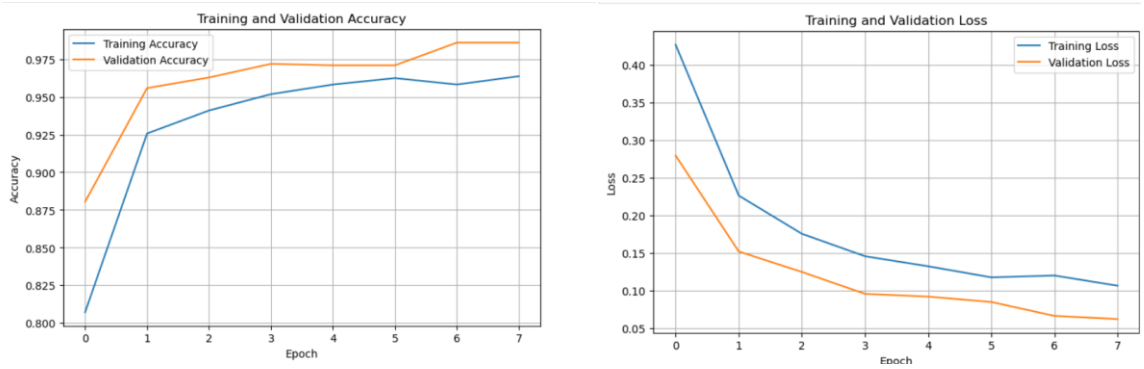


Figure 3. Training and validation accuracy and loss during model training.

4. Decision Threshold Selection

The model produces a probability representing how likely an image is to be defective. Instead of directly using the conventional threshold of 0.50, several thresholds were evaluated on the **validation set**.

Because false negatives are more costly in industrial inspection, a recall-oriented F2 criterion was used. A threshold of **0.30** was selected.

At this threshold, the validation defective recall was **99.47%**, with only 3 false negatives.

The selected threshold was then applied **once to the unseen test set**.

5. Results and Evaluation

Using the selected threshold of 0.30, the model achieved the following test performance:

Metric	Result
Accuracy	**96.92%**
Defective Precision	**95.56%**
Defective Recall	**99.78%**
Defective F1-score	**97.62%**
Macro F1-score	**96.63%**
False Negatives	**1**
False Positives	**21**

The confusion matrix was:

	Predicted OK	Predicted Defective
Actual OK	241	21
Actual Defective	1	452

Figure 4 — Confusion Matrix

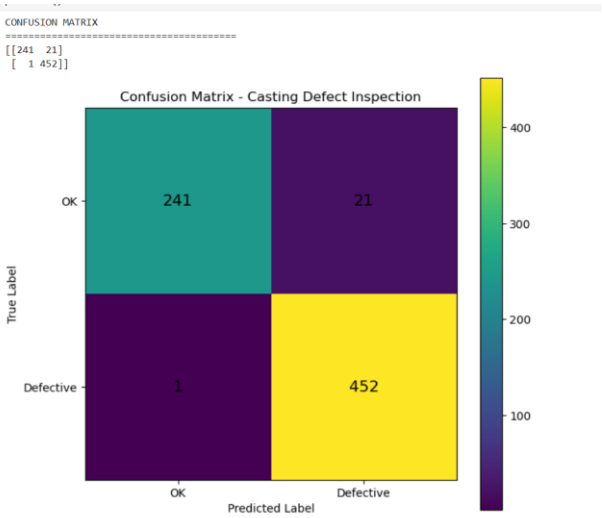


Figure 4. Confusion matrix obtained on the unseen test dataset.

The most important result is the **99.78% recall for defective castings**, meaning that 452 of the 453 actual defective castings were detected.

The threshold comparison also showed the trade-off clearly. At threshold 0.50, there were 6 false negatives, whereas threshold 0.30 reduced this to only **1 false negative**, at the cost of increasing false positives from 6 to 21.

6. Error Analysis

A total of **22 test images were misclassified**: 1 false negative and 21 false positives.

Four representative errors were visualized for analysis.

Figure 5 — Misclassified Test Images

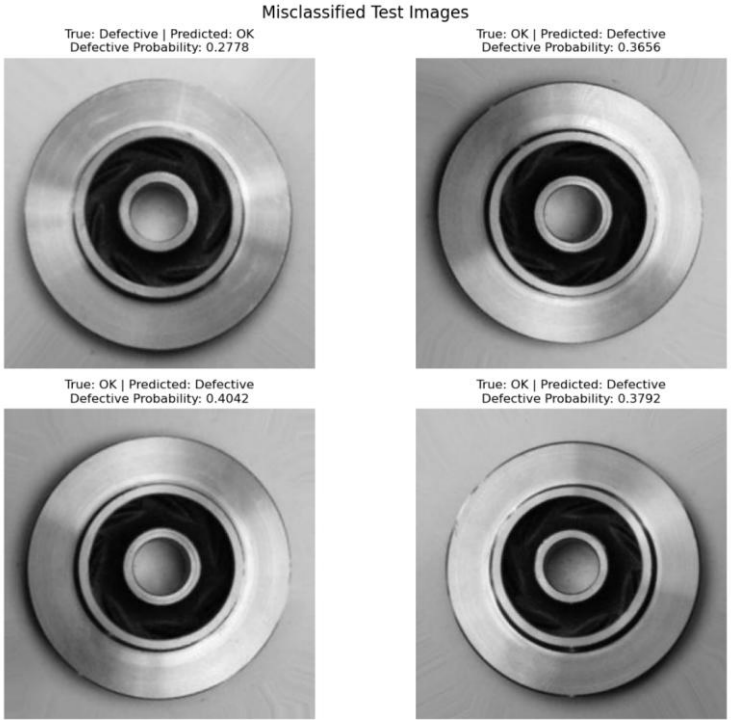


Figure 5. Representative false-negative and false-positive predictions.

The false negative had a defective probability of **0.2778**, which was just below the selected threshold of 0.30. The defect was not visually prominent in the displayed top-view image, making it difficult to distinguish from normal surface variation.

The three selected false positives had probabilities of **0.3656, 0.4042, and 0.3792**. Their surface and structural patterns appeared similar to those of defective samples. These errors suggest that subtle texture, illumination, and surface appearance can overlap between the two classes.

7. Industrial Impact and Limitations

The proposed system can assist quality-control personnel by automatically screening casting images and identifying potentially defective components for further inspection. The high defective recall is particularly useful because missed defects are more costly than false alarms.

However, the model should be considered an **inspection-assistance system rather than a complete replacement for human quality control**. False positives still require manual checking, and performance may vary with changes in camera position, lighting, casting surface condition, or manufacturing processes. Further validation with production-line images would be required before deployment.

8. Conclusion

A MobileNetV2-based transfer-learning classifier was successfully developed for binary casting defect inspection. The model achieved **96.92% test accuracy and 99.78% defective recall** using a decision threshold of 0.30. The threshold was deliberately lowered to reduce missed defective castings, resulting in only **1 false negative out of 453 defective test images**. The results demonstrate that deep-learning-based visual inspection can provide an effective screening mechanism for industrial casting quality control.

9. References

1. **21CSC406J Challenge-Based Learning — Deep Learning Image Classification Problem Statements**, PS-32: Casting Defect Inspection.
2. Ravi Raj Singh, *Real-Life Industrial Dataset of Casting Product*, Kaggle.
3. M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.